[halftone photographic plate — not redrawn]

RENÉ DESCARTES (1596–1650)
Courtesy of *Scripta Mathematica*.

ANALYTIC GEOMETRY

How to read the figures. —— given --- auxiliary (added in the proof)
—— what is being proved or asked

15–1 SOME HISTORY

We have commented before on the fact that numbers, other than integers, did not occur explicitly in Greek geometry. One reason for this was that numbers were hard to work with at that time. Even after the introduction, during the Middle Ages, of Arabic numerals and the place value of the decimal system, geometry continued in the Greek spirit. As a result, it often happened that problems that we would handle naturally with algebra were solved by geometric constructions.

Of course, as time passed, there was a gradual tendency toward the use of algebra in geometric problems, but there was nothing systematic in its use. And until there was sufficient algebraic machinery available and people became skilled in using numbers, there was little advantage in changing a geometric problem into an algebraic problem.

By the beginning of the 17th century, considerable progress had been made in algebra and in algebraic notation;* in particular, quadratic, cubic, and quartic equations could be solved.

Mathematicians were interested in studying tangents to curves, which is more easily done with a little analytic geometry. So it was that in the 17th century the times were right for something new. Indeed, many men had actually been using some analytic geometry without making it a definite system for attacking problems in geometry. Today, looking back on those times, it is hard to imagine why some ideas were so slow in coming. But we must remember that discovery is always much harder than understanding what has been discovered.

It is universally agreed that the French philosopher René Descartes (1596–1650) was the discoverer (inventor?) of analytic geometry, and the date is generally set as 1637, when his *Geometry* was published. Descartes is also noted for his work in physics and in philosophy. In the latter, he aspired to write a philosophy that would be perfectly logical and free from objectionable assumptions. His efforts in philosophy include the book *Discourse on Method*, which expounds what is referred to today as Cartesian philosophy, and which has been a source of dispute ever since! But his *Geometry* is undisputed. Analytic geometry is today a part of a liberal education as well as a necessity for pure and applied science.

Some brief remarks about Descartes' life may be of interest. Born in 1596, he started school in a Jesuit institution at the age of eight. Never a strong, healthy boy, he was allowed by his school to stay in bed mornings to do his studies, and later in life he did much of his writing in bed. He left school in 1612, studied law, and even spent some time in the Dutch army as a gentleman volunteer, leaving in 1619. It appears that by 1620 most of his great ideas had been established in outline. In other words, he had thought them over and had a fairly clear picture

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*The beginning student hardly realizes how much easier the shorthand notation of algebra makes his problems. The interested student is urged to read in the history of mathematics and to try working a problem in earlier notations.

of how they should go. Much of his life thereafter was spent in Holland. A devout man, he nevertheless had troubles with the church over his philosophical ideas. By the time his *Geometry* was published, he was a well-known figure. In 1649 he visited the court of Queen Christina of Sweden, after much urging, to teach her his system of philosophy. Assigned the outlandish hour of 5 a.m. three days a week for the instructions, he fell ill in February 1650 and died a few days later.

Geometry was his only book on mathematics and, in a sense, contained nothing totally new. What Descartes did was to show how a systematic use of coordinates helps to attack problems. He classified various kinds of curves and suggested further work. The book was suggestive and tremendously fruitful. We owe to him our convention of using the first letters of the alphabet (a, b, c) for knowns and the last letters (x, y, z) for unknowns.

It is not easy to give a concise definition of analytic geometry. In a sense, this entire chapter is such a definition, for a complete definition should include some of its uses. But for the moment, it may suffice to say that, in analytic geometry, geometric figures are studied by means of equations. Then, if we wish to know something concerning the geometric figure, it *may* be easy to learn this by examining an equation. The equations that are used involve numbers called coordinates, which serve to locate points. But why the word "analytic"? Analytic geometry does enable one to analyze geometric figures, but this is also done in regular Euclidean geometry. Perhaps a better description would be to call this new way of looking at geometry, plane *coordinate* geometry, or plane *Cartesian* geometry. Remember that it is the same plane we are studying all the time, no matter how we look at it.

Exercise Group 15–1

1. Read the history of mathematics during the 17th century.
2. Read a biography of Descartes in an encyclopedia or in *Men of Mathematics*, by E. T. Bell.

15–2 EXAMPLES

Although analytic geometry is distinguished by use of coordinates, the idea should already be familiar from some common situations. Some cities are laid out in squares, with the east-west and north-south roads called avenues and streets, respectively. The numbers of street and avenue fix any intersection. In the same way, you unconsciously use coordinates in describing a nearby town as, perhaps, 7 miles north and 5 miles west of "here."

Exercise Group 15–2

1. Try to think of other examples of coordinates, that is situations in which position is determined by a pair of numbers.
2. Discuss the need to have streets straight in order to "coordinatize" a city.

By far the most important example of coordinates is our method (Chapter 5) of putting coordinates on a line. We recall how this was done:

Given a line l and a unit of length, choose a point O on the line l, as in Fig. 15–1. Now choose* a point P_1 on l at a distance 1 from O. This choice determines which half of the line we will call the *positive axis*, or *positive direction*. For every positive real number x, there is exactly one point P_x of l on the same side of O as the point P_1 such that the length $\overline{OP_x} = x$. For every negative real number x', there is exactly one point $P_{x'}$ of l on the opposite side of O from P_1 such that the length $\overline{OP_{x'}} = -x' = |x'|$.

Each of the numbers x and x' is called the *coordinate* of the corresponding point of the line.

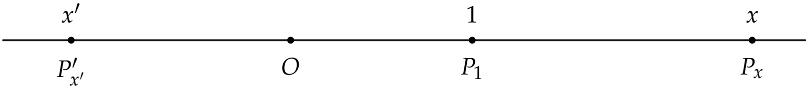

FIGURE 15–1

Exercise Group 15–3

1. How are the points P_{x_1}, P_{x_2}, and O related if $x_1 < 0$ and $x_2 > 0$? if $0 < x_2 < x_1$? if $x_1 < x_2 < 0$?

2. If we agree that points with larger coordinates "lie to the right" of points with lesser coordinates, how are we thinking of the drawn line l and the chosen point P_1?

3. Two points on the line have coordinates -3 and 7. What is the distance between them? between 5 and 12? between x_1 and x_2?

Definition 15–1. We will refer to the above construction of coordinates as a coordinatization of the line l, with the point O as *origin of coordinates*.

15–3 THE EXISTENCE OF COORDINATES

The construction of plane coordinates is best carried out in a number of simple steps. We assume that a unit of length has been selected in the plane.

*There are two ways of choosing this point, namely on either side of O. Which side is chosen is not important.

Definition 15–2. Choose a point O in the plane. Call this point the *origin of coordinates*. Choose any two perpendicular lines through O. Call these lines *coordinate axes*. Coordinatize each of the axes, with the point O as origin of coordinates. Select one of these lines and call it the *axis of abscissas*.* The other axis is called the *axis of ordinates*.† See Fig. 15–2.

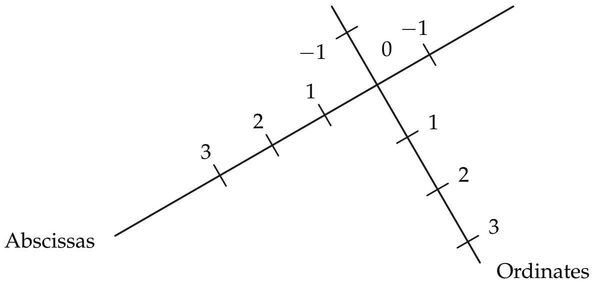

FIGURE 15–2

Exercise Group 15–4

1. Draw a figure, different from Fig. 15–2, illustrating Definition 15–2.
2. Compare your figure with your neighbor's. Did you both choose the "same" axis for abscissas? Turn your figure sideways. Do

*This is not a fortunately chosen word. It would be better to say "first coordinates." In most drawings in textbooks, the axis of abscissas is drawn horizontally as the reader views it. There is nothing sacred about this, we just agree to consider one coordinate before we do the other (something must be first) and call this first coordinate by the strange name *abscissa*.

† This word is not much better, but the literature of mathematics is filled with this word too, so make the best of it and learn it. Of course "second coordinate" is much better.

you have a different opinion about which is the first coordinate?

3. With the same drawing as in Exercise 1, reverse the coordinates on each of the axes. Is this a possible way of satisfying Definition 15–2? Suppose that we reversed the coordinates on only one of the axes.

Perhaps you will have observed a certain arbitrariness in the definition, in that we *choose* a point O and *choose* lines for axes. This is actually the way things operate in practice. In applying analytic geometry to problems in physics or engineering, it is usually necessary to *choose* coordinates. *There are no fixed coordinate systems inherent in nature.* Coordinate systems are convenient inventions that facilitate mathematical computations.

Having set up our axes, we can make certain statements concerning them. We give some of these in a theorem.

Theorem 15–1. *The coordinate axes separate the plane into four regions, called quadrants, such that if P and Q are points not on either axis and in the same quadrant, then the segment PQ does not meet either axis, whereas if P and Q are in different quadrants, then PQ meets at least one of the axes. See Fig. 15–3.*

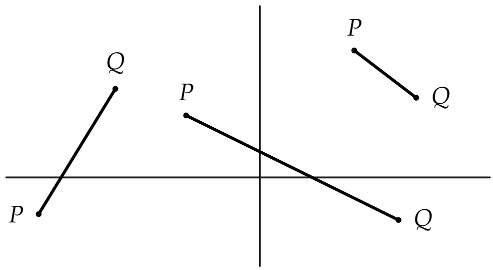

FIGURE 15–3

Proof. Each axis separates the plane. (Why?) Consider any point

P not on any axis. The point P is on one side of the axis of abscissas and one side of the axis of ordinates. Let Q be any point on the same side of the axis of abscissas as P and on the same side of the axis of ordinates as P. Then PQ meets neither axis. (Why?) The set of all such points Q is a quadrant. If Q_1 and Q_2 are two points of that quadrant, then Q_1Q_2 does not intersect either axis. (Why?)

There are four ways of choosing the point P, according to which sides of the axes it is on. This gives rise to the four quadrants.

Exercise Group 15–5

1. Describe the four ways of choosing P to get the four quadrants.
2. Does the above proof require the axes to be perpendicular?

Definition 15–3. The half plane that contains the positive axis of abscissas will be called the *right half plane*, and similarly for *left half plane*.* The half plane that contains the positive axis of ordinates will be called the *upper half plane*, and similarly for *lower half plane*.

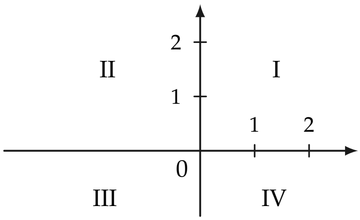

FIGURE 15–4

*This agrees with the way axes are conventionally drawn in textbooks.

The quadrant that is in the right half plane and in the upper half plane will be called the *first quadrant;* the *second quadrant* lies in the upper and left half planes; the *third quadrant*, in the lower and left half planes; and the *fourth quadrant*, in the lower and right half planes. See Fig. 15–4.

Exercise Group 15–6

1. In Fig. 15–5, which is quadrant I? quadrant II? quadrant III? quadrant IV?

*2. Explain why Definition 15–3 does not require any previous knowledge of what the words "left" and "right" mean.

*3. Has there been any use made of the fact that the axes have been selected to be perpendicular? What changes would have to be made so far if the axes were not at right angles?

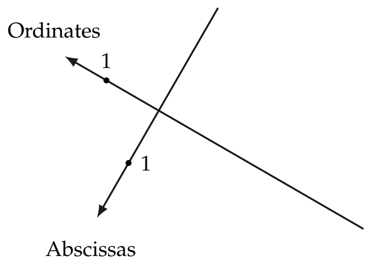

FIGURE 15–5

Definition 15–4. Let P be any point of the plane, and l_1 and l_2 the lines through P parallel, respectively, to the axis of ordinates and axis of abscissas. The line l_1 meets the first axis in a point A, with coordinate x. The line l_2 meets the second axis in a point B, with coordinate y. See Fig. 15–6. The two numbers x and y are called the *coordinates* of P. We write them in parentheses, (x, y), and refer to (x, y) as a *point*. The number x is the *abscissa* of P, and the number y is the *ordinate* of P.

Because the letters x and y are so frequently used for abscissa and ordinate, the axes are often called the *x-axis* and the *y-axis*. The plane then, when coordinatized in this way, is often called the *xy-plane*.†

Given the coordinates of a point, if one marks that point in a figure, it is then said that the point has been *plotted*.

Remark. The point O has coordinates $(0, 0)$. (Why?)

If P does not lie on either axis, neither coordinate of P can be 0. If the first coordinate of P is 0, then P is on the axis of ordinates. The converse is also true.

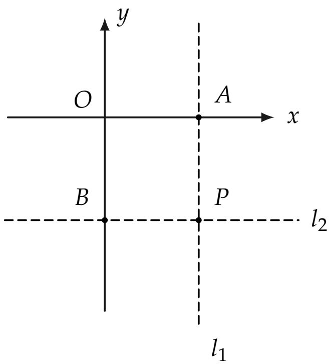

FIGURE 15–6

Exercise Group 15–7

† Of course there is nothing really special about the letters x and y. Use of different letters, say u and v, would give us a *uv-plane*.

1. Draw axes, and plot the points $(2,4)$; $(-2,4)$; $(-3,-\frac{1}{2})$; $(0,-5)$; $(-4,0)$; and $(7\frac{1}{2},-3)$.

2. A point P lies in the first quadrant. The perpendicular distance from the x-axis is $4\frac{1}{2}$ units; the distance from the y-axis is 3 units. What are the coordinates of P? What are the coordinates if distances are the same but P is in the second quadrant?

*3. Suppose that, using the same lines as axes, the unit of distance is chosen twice as large. What would be the effect upon the coordinates of points?

*4. Where are all the points in the plane with abscissa 0? abscissa negative? ordinate positive?

*5. Where are all the points in the plane whose ordinates are equal to their abscissas?

6. Draw the line that passes through the points $(-1,5)$ and $(2,-1)$. Can you easily draw a line parallel to it, which passes through the origin?

*7. Carry out the construction of coordinates when the axes are not perpendicular.

8. Both coordinates of the point S are negative numbers. The points S and T are on the same side of the x-axis, but on opposite sides of the y-axis. What can you say about the coordinates of T?

9. The positive and negative x- and y-axes are rays from O. Describe the four quadrants as interiors of angles formed by these rays.

10. The abscissa of a point is 0. Where must it lie?

11. The ordinate of a point is 0. Where can it be?

12. What can you say about a point if the product of its coordinates is negative? 0? positive?

13. Plot five distinct points with first coordinate equal to 3. Describe the set of all points in the plane such

that the first coordinate of every point of the set is 3.

14. Consider the set of all points in the plane such that the sum of the coordinates of each point is 10. Plot six such points, two in each of quadrants I, II, and IV. Why are there no such points in quadrant III?

15. Describe the set of all points (x, y) such that $x + 1 < 0$.

16. What is the perpendicular distance from $(3, -4)$ to the x-axis? to the y-axis? from $(-5, -7)$ to each axis?

17. How far is $(3, 4)$ from $(0, 0)$?

15–4 STRAIGHT LINES

Having coordinates, we naturally ask how lines may be described by coordinates. In exercises we have already examined a few lines.

For example, the set of all points (x, y) with abscissa 3 is the line shown in Fig. 15–7, and every point on the line parallel to the y-axis, at a distance 3 to the right of the y-axis, has its x-coordinate equal to 3. Hence, this line can be specified briefly by writing just $x = 3$. We say then that "$x = 3$" is the *equation* of this line.

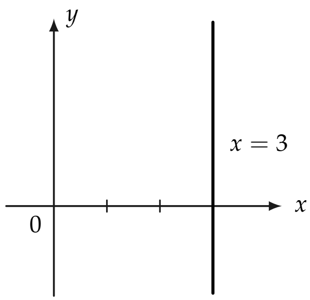

FIGURE 15–7

Conversely, when we see the equation $x = 3$, we ask ourselves, "What is the set of all points (x, y) for which $x = 3$?"

In the same way, all lines parallel to the y-axis have an equation $x = a$, where a is some real number, positive or negative. (If $a = 0$, what is the line?)

Likewise, lines parallel to the x-axis will have equations $y = b$.

Exercise Group 15–8

1. A line is parallel to the y-axis and passes through the point $(-3, 5)$. What is its equation? What if it is parallel to the x-axis?

2. What is the equation of the line that bisects the first and third quadrants? the second and fourth?

*3. Plot several points (x, y) whose coordinates satisfy the equation $x + y = 5$. Can you describe briefly the set of all points (x, y) whose coordinates satisfy this equation?

We will prove shortly that every straight line has an equation of a simple kind. But first we see that such simple equations turn out to be straight lines.

Example 1. Consider the set of all points (x, y) whose coordinates satisfy the equation $2x - 3y + 6 = 0$. We first find a few of these points, by choosing a value for x (or y) and then solving for y (or x). Thus, if $x = 0$, we have $0 - 3y + 6 = 0$, or $y = 2$. If $x = 3$, then $y = 4$. If $y = 0$, then $x = -3$.

x	y
0	2
3	4
−3	0

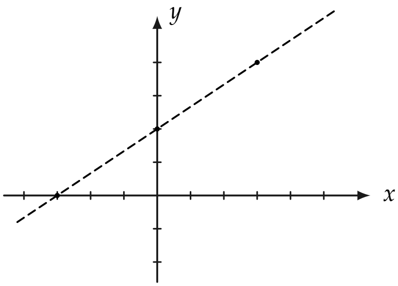

FIGURE 15–8

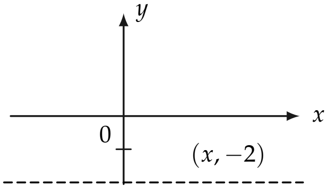

FIGURE 15–9

When these points are plotted, we see (by looking at Fig. 15–8) that they apparently lie on a line.

Find other points whose coordinates satisfy the equation, and plot them.

Example 2. Consider the set of all points (x, y) whose coordinates satisfy the equation $-4y - 8 = 0$. From the equation, we see that $y = -2$, regardless of what x is. The set of all points $(x, -2)$ is the line parallel to the x-axis and a distance 2 units below. See Fig. 15–9.

Exercise Group 15–9

Plot several points (x, y) whose coordinates satisfy the following equations:

1. $2x + 3y - 6 = 0$.
2. $x - 2y + 4 = 0$.
3. $3x - 9 = 0$.
4. $-x - y + 4 = 0$.
5. $2ax + 3ay - 6a = 0$; $a \neq 0$.
6. $-3x + 5y - 15 = 0$.
7. $y = 2x + 2$.
8. $x = 3y - 6$.

From the examples we observe that equations of the form $Ax + By + C = 0$, where A, B, and C are given numbers, are satisfied by the coordinates of points on lines. We now prove this theorem.

***Theorem 15–2.** *If A, B, and C are given numbers, with not both A and B zero, then all points (x, y) whose coordinates satisfy the equation $Ax + By + C = 0$ lie on a single line.*

Proof. We separate the proof into three cases.

Case 1. $A = 0$. Then $B \neq 0$ (why?) and the equation is $By + C = 0$, or $y = -C/B$. Then all points with this y-coordinate lie on the line parallel to the x-axis and passing through $(0, -C/B)$.

Case 2. $B = 0$. In this case, the points lie on a line parallel to the y-axis.

Case 3. $A \neq 0$ and $B \neq 0$. It suffices to show that any two points (x_1, y_1) and (x_2, y_2) whose coordinates satisfy the equation lie on one line with the point $(0, -C/B)$. This will be the case if and only if the right triangles indicated in Fig. 15–10 are similar, where

$$y_1 = \frac{-Ax_1 - C}{B}; \qquad y_2 = \frac{-Ax_2 - C}{B}. \tag{15–1}$$

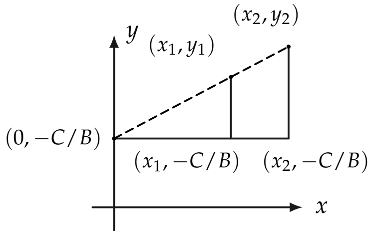

FIGURE 15–10

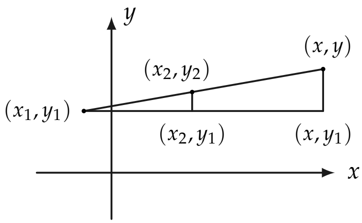

FIGURE 15–11

But these triangles are similar if and only if

$$\frac{y_2 - (-C/B)}{x_2} = \frac{y_1 - (-C/B)}{x_1}. \tag{15–2}$$

From Eq. (15–1) we see that

$$\frac{y_2 + C/B}{x_2} = \frac{-A}{B} = \frac{y_1 + C/B}{x_1}.$$

Hence we conclude that the triangles are similar and the points $[0, -(C/B)]$, (x_1, y_1), and (x_2, y_2) are collinear.

***Theorem 15–3.** *If l is a line, then there are numbers A, B, and C, with not both A and B zero, such that every point (x, y) on l has coordinates that satisfy the equation*

$$Ax + By + C = 0. \tag{15–3}$$

Proof. As in the previous theorem, we consider three cases.

Case 1. l is parallel to the x-axis. Then there is a point $(0, b)$ on l, and every point (x, y) on l satisfies $y = b$. This is of the form of Eq. (15–3), with $A = 0$, $B = 1$, and $C = -b$.

Case 2. l is parallel to the y-axis. This case is left to the student.

Case 3. l is parallel to neither axis. Choose (x_1, y_1) and (x_2, y_2), distinct points on l. Then $x_1 \neq x_2$ and $y_1 \neq y_2$. (Why?) Let (x, y) be any other point on l.

As shown in Fig. 15–11, the points (x_2, y_1) and (x, y_1) form, with (x_1, y_1), (x_2, y_2), and (x, y), two right triangles that are similar. Because of similarity,†

$$\frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1} \tag{15–4}$$

or

$$(y_2 - y_1)x - (x_2 - x_1)y + x_2 y_1 - y_2 x_1 = 0,$$

which is of the form of Eq. (15–3). This completes the proof.

† Figure 15–11 is drawn so that $x_1 < x_2 < x$, and $y_1 < y_2 < y$, but all orders can occur for different points and different lines. In every case, however, Eq. (15–4) will be satisfied. Thus it makes no difference which point is considered as (x_1, y_1), so long as one is consistent in its use.

Remark 1. An equation equivalent to Eq. (15–4) is

$$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}\,(x - x_1). \tag{15–5}$$

This is a useful equation to remember because it enables one to write at once an equation satisfied by the coordinates of any point on a line through two given points.

Remark 2. Two different equations may correspond to the same line. Thus for example, $x = 2$ and $2x - 4 = 0$ represent the same line. Such equations are said to be *equivalent* equations. In arriving at equations for lines, we usually do not care which equivalent equation we obtain.

Definition 15–5. If $Ax + By + C = 0$ is an equation satisfied by the coordinates (x, y) of all points on a line l, we say that $Ax + By + C = 0$ is the equation of l. We also speak of the *"line $Ax + By + C = 0$."* Such an equation of the first degree in x and y will be called a *linear equation* in x and y.

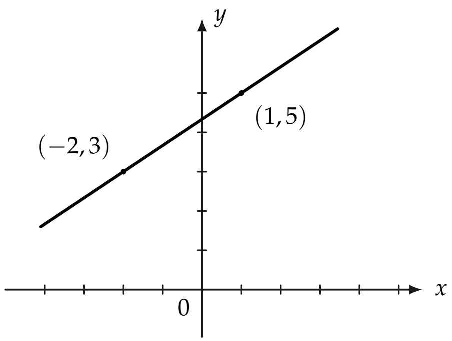

FIGURE 15–12

Example. Find the equation of the line through $(-2, 3)$ and $(1, 5)$

in Fig. 15–12, and draw the line. Using Eq. (15–5), we have

$$y-3=\frac{5-3}{1-(-2)}\,[x-(-2)] \qquad \text{or} \qquad y=\tfrac{2}{3}x+\tfrac{13}{3}.$$

Exercise Group 15–10

1. Draw the lines

   (a) $3x-4y=12$,
   (b) $3x-4y=8$,
   (c) $4x+3y=0$,
   (d) $4x+15=0$,
   (e) $-2x-3y+5=0$,
   (f) $y=\dfrac{-1}{2}x+3$,
   (g) $y=\dfrac{-1}{2}x$,
   (h) $x=3y+3$,
   (k) $5y-12=0$,
   (l) $2x+y=7$.

2. Find the equations of the lines through the following pairs of points:

   (a) $(-1,1)$ and $(3,-1)$,
   (b) $(2,3)$ and $(-2,-3)$,
   (c) $(a,0)$ and $(0,b)$,
   (d) $(-3,0)$ and $(0,-5)$,
   (e) $(-2,-4)$ and $(-5,-5)$,
   (f) $(0,b)$ and $(1,m+b)$.

3. In Exercise 1, how many points did you plot to draw each line? Is there a reason for plotting more than two?

4. Find the coordinates of the point of intersection of the lines $3x-5y=7$ and $2x+3y=-1$.

5. A line meets the y-axis in $(0,-3)$ and the x-axis in $(-4,0)$. Where does this line intersect the line $y=7x-19$?

6. Show that the lines $2x+3y=0$ and $4x+6y=5$ are parallel.

15–5 DISTANCE

The distance between any two points of the plane is easy to calculate from the coordinates of the points. It is particularly easy if the line through the points is parallel to one of the axes. Thus, $\overline{AB} = 1 - (-3) = 4$, and $\overline{CD} = |x_2 - x_1| = |x_1 - x_2|$ in Fig. 15–13.

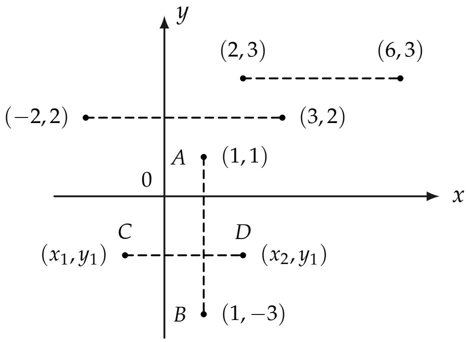

FIGURE 15–13

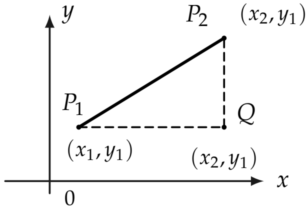

FIGURE 15–14

If the line through the points is not parallel to one of the axes, then with the point $Q\ (x_2, y_1)$ a right triangle P_1P_2Q is formed, as in Fig. 15–14, and the Pythagorean theorem asserts that $\overline{P_1P_2}^2 = \overline{P_1Q}^2 + \overline{P_2Q}^2 = |x_2 - x_1|^2 + |y_2 - y_1|^2$. Taking square roots, we obtain $\overline{P_1P_2} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.

Remark. In using this formula it makes no difference which point is considered to be (x_1, y_1) because $(x_1 - x_2)^2 = (x_2 - x_1)^2$ and $(y_1 - y_2)^2 = (y_2 - y_1)^2$.

Exercise Group 15–11

1. Show that the triangle with vertices $(2,6)$, $(5,7)$, and $(-8,-2)$ is a right triangle.

2. What kind of triangle is the one with vertices $(3,2)$, $(-4,4)$, and $(1,-5)$?

*3. Write an equation that states that the point (x, y)

is equidistant from $(-3, 2)$ and $(2, 1)$. Show that this equation is equivalent to a linear equation. Hence conclude that the set of all points equidistant from $(-3, 2)$ and $(2, 1)$ is a straight line.

4. Show that $(-1, -18)$, $(-13, -2)$, $(11, -8)$, and $(-1, 8)$ are vertices of a parallelogram.

5. Show that if (x, y) is a point on a circle whose center is $(-2, 1)$ and radius is 3, then $x^2 + y^2 + 4x - 2y = 4$.

6. Show that the points $(2, 1)$, $(3, \frac{1}{2})$, and $(-2, 3)$ lie on a line. What geometric theorems do you use?

7. Show that the distance formula yields $|x_1 - x_2|$ if $y_1 = y_2$.

15–6 MIDPOINT

The midpoint of the segment P_1P_2 may be shown to have coordinates

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right).$$

The proof is left to the student. If the line through the points is parallel to one of the axes, the proof is very simple. If $x_1 \neq x_2$ and $y_1 \neq y_2$, a right triangle may be formed as in Fig. 15–15, and parallels to the coordinate axes through the midpoint will bisect the sides in points whose coordinates are easily computed.

Example. Show in Fig. 15–16 that the midpoint formula actually gives a point equidistant from $(-1, 3)$ and $(1, -1)$.

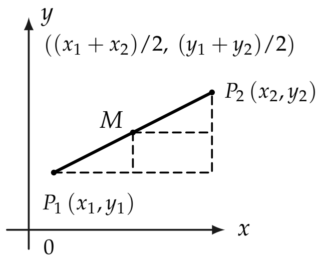

FIGURE 15–15

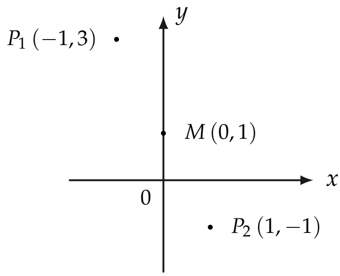

FIGURE 15–16

The point M is $\left(\dfrac{-1+1}{2}, \dfrac{3+(-1)}{2}\right) = (0,1)$. Then

$$\overline{P_1P_2} = \sqrt{2^2+4^2} = \sqrt{20} = 2\sqrt{5},$$

$$\overline{P_1M} = \sqrt{1^2+2^2} = \sqrt{5},$$

and

$$\overline{MP_2} = \sqrt{1^2+2^2} = \sqrt{5}.$$

Hence

$$\overline{P_1P_2} = 2\sqrt{5} = \sqrt{5}+\sqrt{5} = \overline{P_1M}+\overline{MP_2}.$$

Exercise Group 15–12

1. Find the midpoints of the sides, and the lengths of the medians, of the triangle whose vertices are $(-4, 10)$, $(12, 10)$, and $(4, -2)$.

2. Find the coordinates of the points that divide the segment $(-3, -7)$ $(11, 17)$ into fourths.

*3. Find the points that trisect the segment $(-3, -7)$ $(6, 11)$.

*4. Derive a formula for the coordinates of the points that trisect the segment (x_1, y_1)

(x_2, y_2).

*5. Show that the medians of the triangle with vertices (x_1, y_1), (x_2, y_2), and (x_3, y_3) intersect in the point $[(x_1 + x_2 + x_3)/3, (y_1 + y_2 + y_3)/3]$.

6. The point $(-3, 4)$ is the midpoint of a segment, one end of which is $(5, 7)$. Find the other end.

*7. In Fig. 15–17, show that the line joining the midpoints of the sides of a triangle is half as long as the third side, and parallel to it. [*Hint:* Choose your coordinate system so that one vertex is at the origin and another is on the positive x-axis.]

*8. Prove that the diagonals of a parallelogram bisect each other.

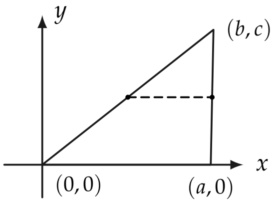

FIGURE 15–17

15–7 CURVES AND EQUATIONS

We have seen that certain sets of points can be described by equations. Thus a straight line has a linear or first degree equation, in the sense that the coordinates of every point on the line satisfy an equation of first degree in x and y of the form $Ax + By + C = 0$, and conversely. In the same way, other sets of points can be described by different equations. Perhaps the simplest example of this is furnished by circles.

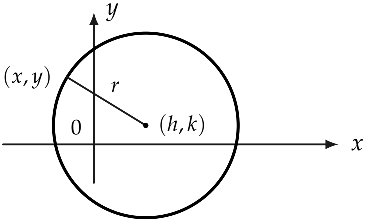

FIGURE 15–18

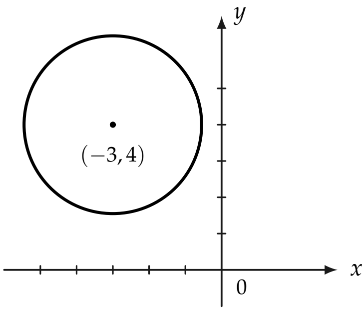

FIGURE 15–19

Suppose that (x, y) in Fig. 15–18 is a point on the circle whose center is (h, k) and whose radius is r, then

$$\sqrt{(x-h)^2 + (y-k)^2} = r$$

or, equivalently,

$$(x-h)^2 + (y-k)^2 = r^2. \tag{15–6}$$

Conversely, a point (x, y) whose coordinates satisfy Eq. (15–6) will be at a distance r from (h, k) and therefore lies on the circle.

We speak of Eq. (15–6) as "the" equation of a circle, although any equivalent equation would also be called "the" equation. For example, the equation of the circle whose center is at $(-2, 3)$, and whose radius is 4, is

$$(x+2)^2 + (y-3)^2 = 16$$

or

$$x^2 + y^2 + 4x - 6y = 3.$$

Given an equation in the form of Eq. (15–6), we can recognize it at once for the equation of a circle. Any equation equivalent to Eq. (15–6) is the equation of a circle. For example, consider the equation

$$x^2 + y^2 + 6x - 8y + 19 = 0.$$

Rewrite the equation as

$$(x^2 + 6x + ?) + (y^2 - 8y + ?) = -19 + ? + ?.$$

By completing the squares of the expressions in parentheses, we have

$$(x^2 + 6x + 9) + (y^2 - 8y + 16) = -19 + 9 + 16$$

or

$$(x+3)^2 + (y-4)^2 = 6,$$

which is the equation of a circle whose center is at $(-3, 4)$ and whose radius is $\sqrt{6}$. See Fig. 15–19.

Other curves (sets of points) will have other kinds of equations. The student is referred to books on analytic geometry for more detail.

Exercise Group 15–13

1. Write the equation of the circle whose center is $(3, 5)$ and whose radius is 6.

2. Write the equation of the circle whose center is at $(-4, -3)$ and which (a) is tangent to the y-axis, (b) passes through the origin.

3. Find center and radius of the following circles:

   (a) $x^2 + y^2 + 8x - 9 = 0$,
   (b) $x^2 + y^2 - 6x - 10y = 2$,
   (c) $x^2 + y^2 - 7 = 0$,
   (d) $2x^2 + 2y^2 + 5x + 4y - 5 = 0$,
   (e) $x^2 + y^2 - 6x - 8y + 25 = 0$.

*4. Find an equation satisfied by the coordinates of every point (x, y) whose distance from $(2, 0)$ is equal to its distance from the line whose equation is $x + 2 = 0$.

*5. If the sum of the distances from (x, y) to $(-3, 0)$ and from (x, y) to $(3, 0)$ is equal to 10, show that $16x^2 +$

$25y^2 = 400.$